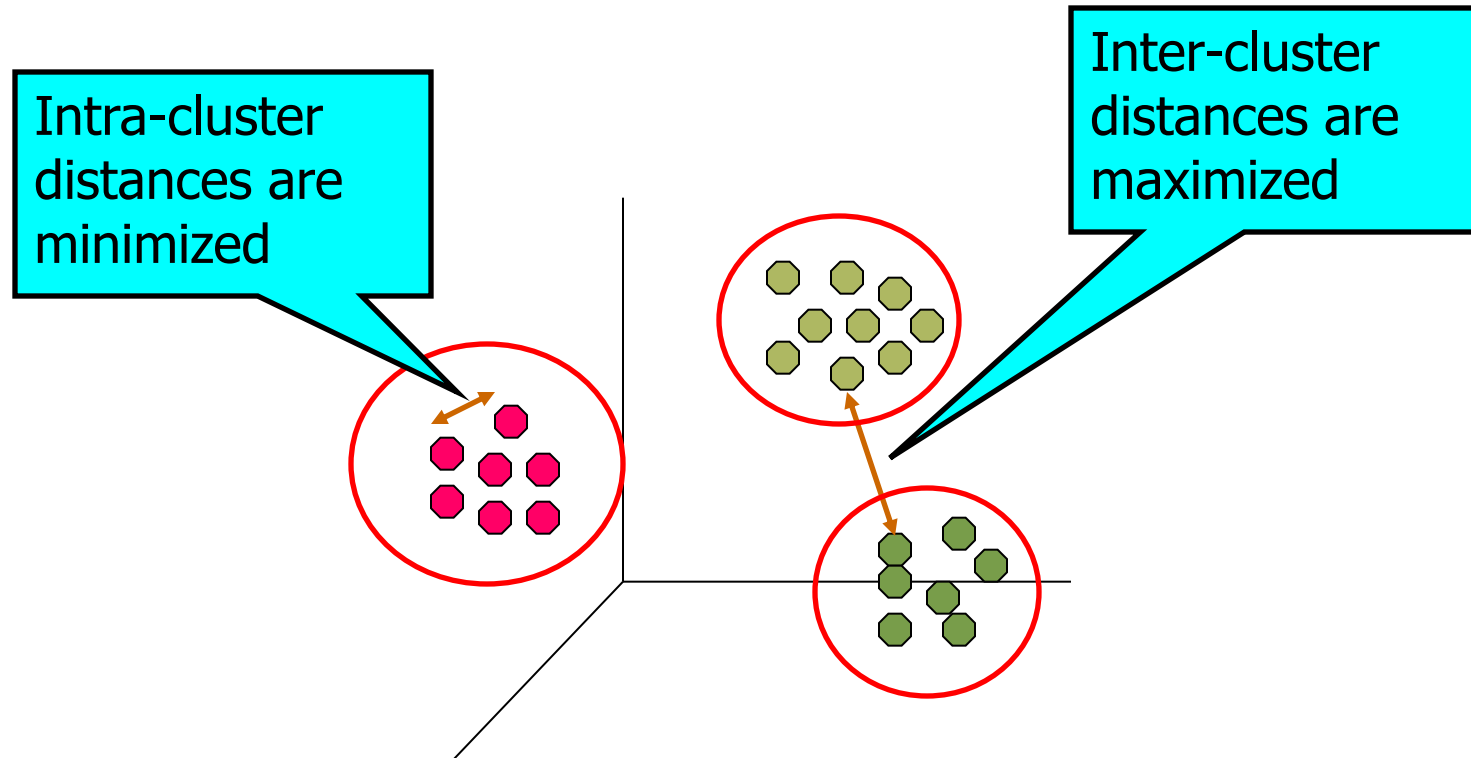


Cluster Analysis

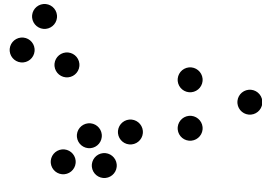
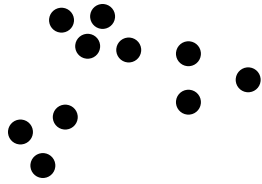
Serkan Ayvaz

What is Cluster Analysis?

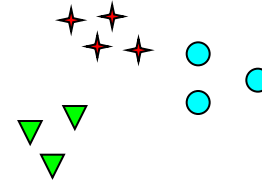
Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



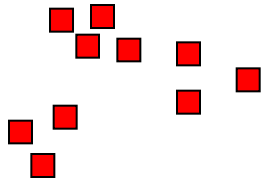
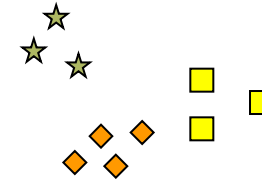
Notion of a Cluster can be Ambiguous



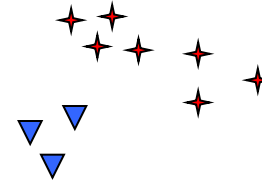
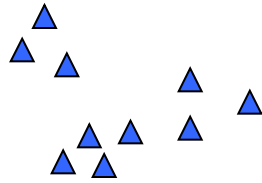
How many clusters?



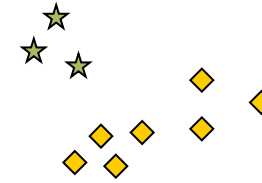
Six Clusters



Two Clusters



Four Clusters



Types of Clusterings

A clustering is a set of clusters

Important distinction between hierarchical and partitional sets of clusters

Partitional Clustering

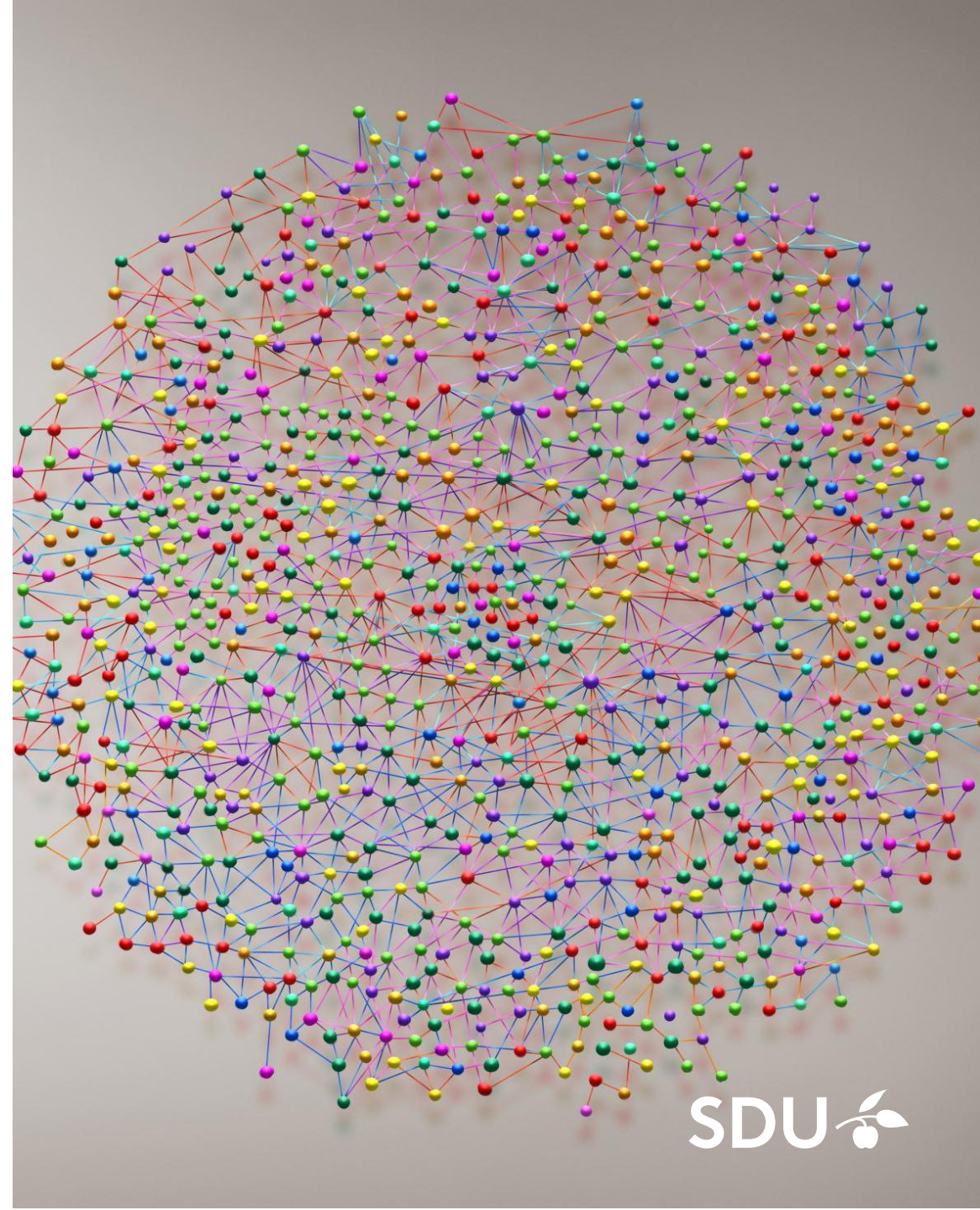
- A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset

Hierarchical clustering

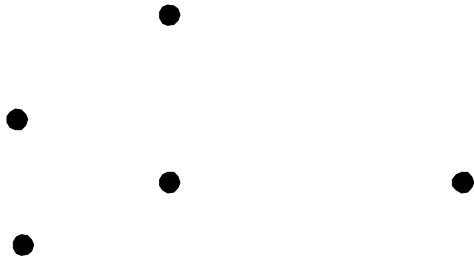
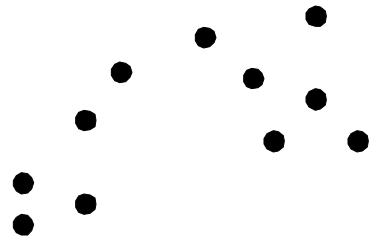
- A set of nested clusters organized as a hierarchical tree

Density based clustering

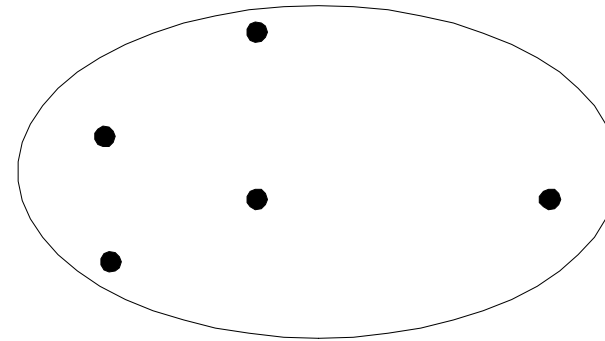
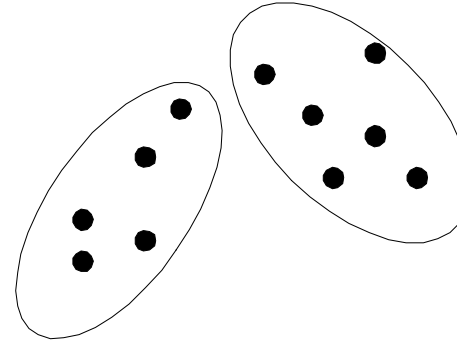
- A set of clusters created based on density of data points



Partitional Clustering

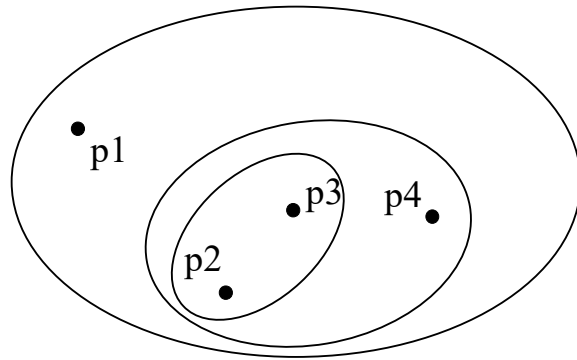


Original Points

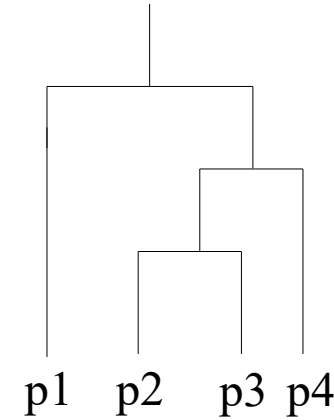


A Partitional Clustering

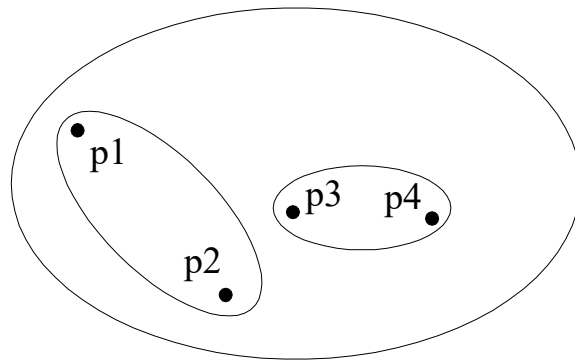
Hierarchical Clustering



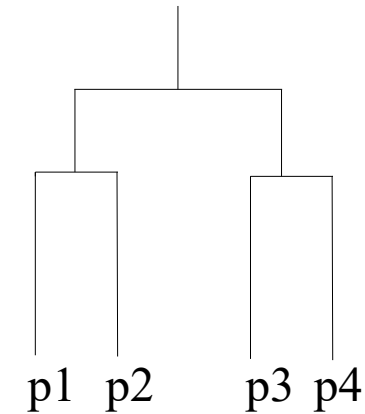
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

Clustering Algorithms



Partitional Clustering

K-means and its variants



Hierarchical clustering

Agglomerative(Bottom-up) clustering
Top-Down clustering



Density Based clustering

DB-SCAN

K-means Clustering

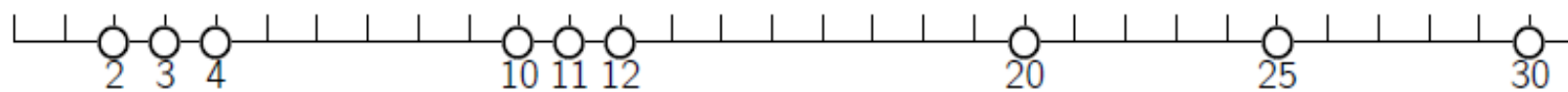
Partitional clustering approach

- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid

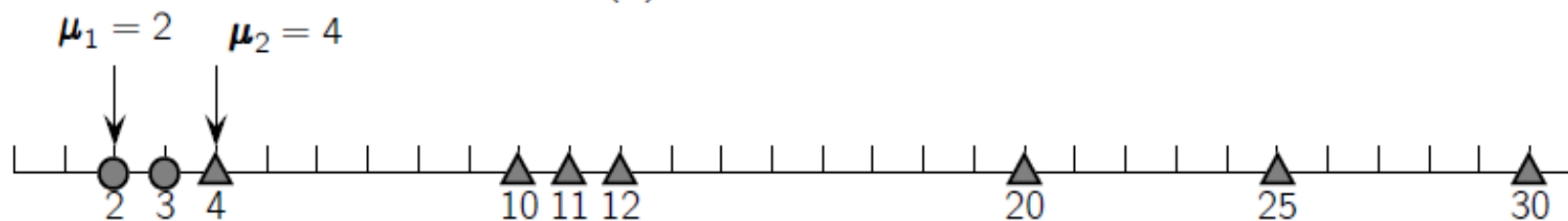
Number of clusters, K , must be specified

The basic algorithm is very simple

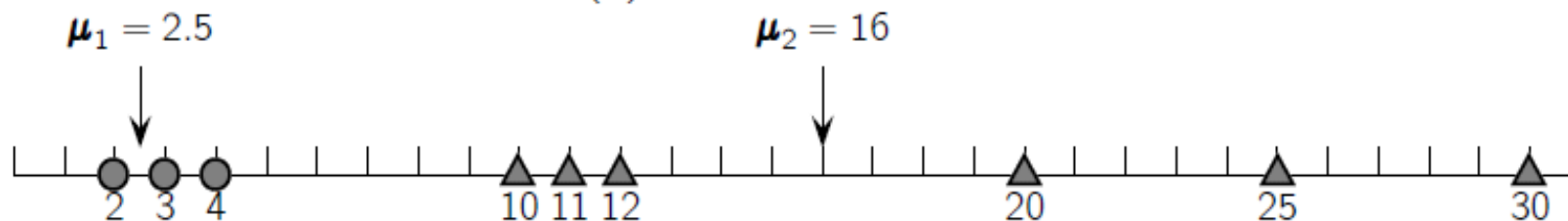
-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-



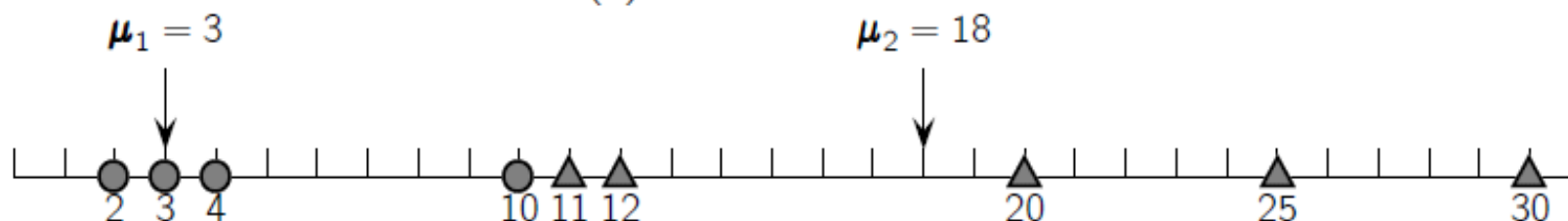
(a) Initial dataset



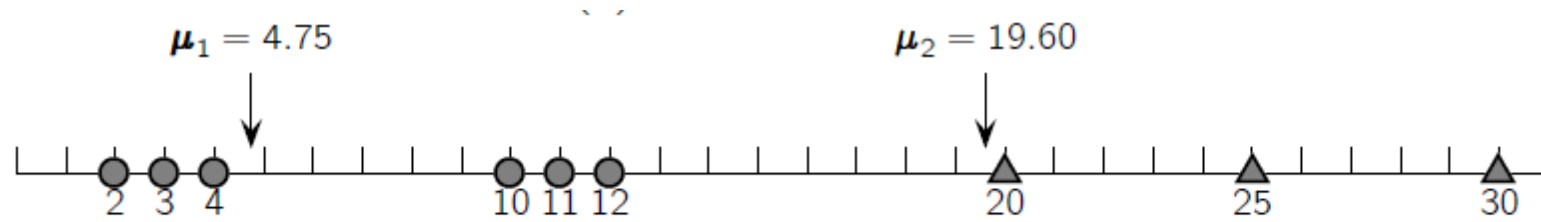
(b) Iteration: $t = 1$



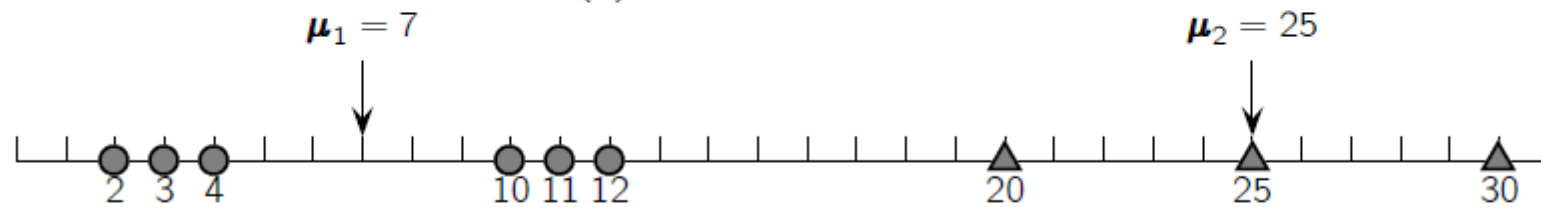
(c) Iteration: $t = 2$



(d) Iteration: $t = 3$



(e) Iteration: $t = 4$



(f) Iteration: $t = 5$ (converged)

Issues and Limitations for K-means

How to choose initial centers?

How to choose K?

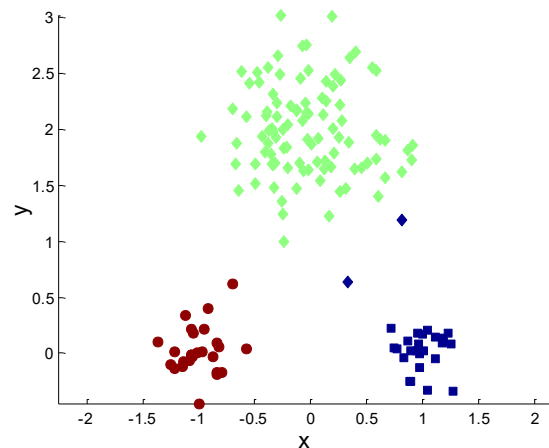
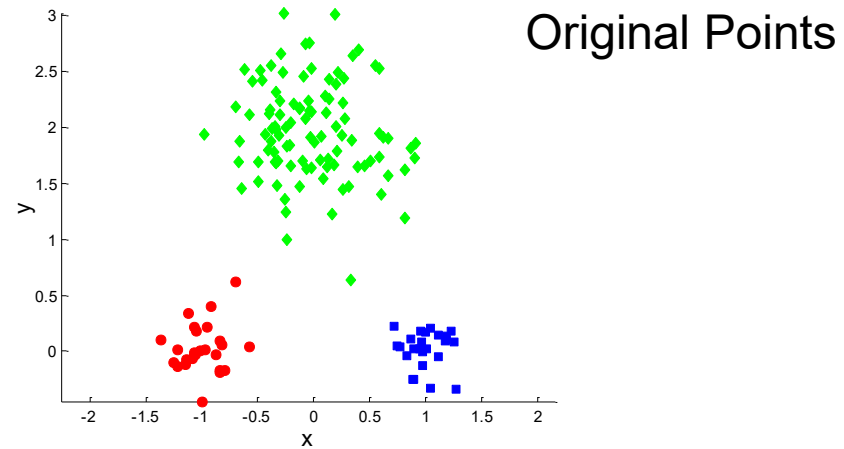
How to handle Outliers?

Clusters different in

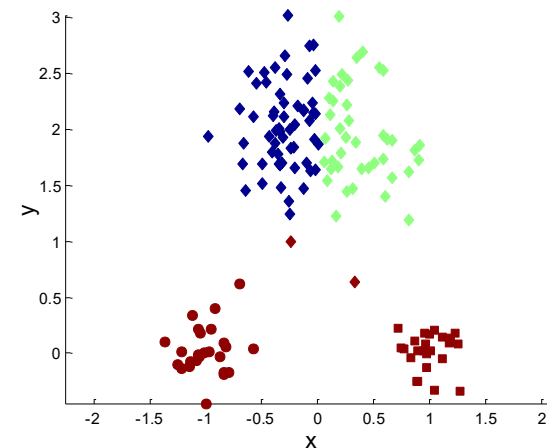
- Shape
- Density
- Size



Two different K-means Clusterings

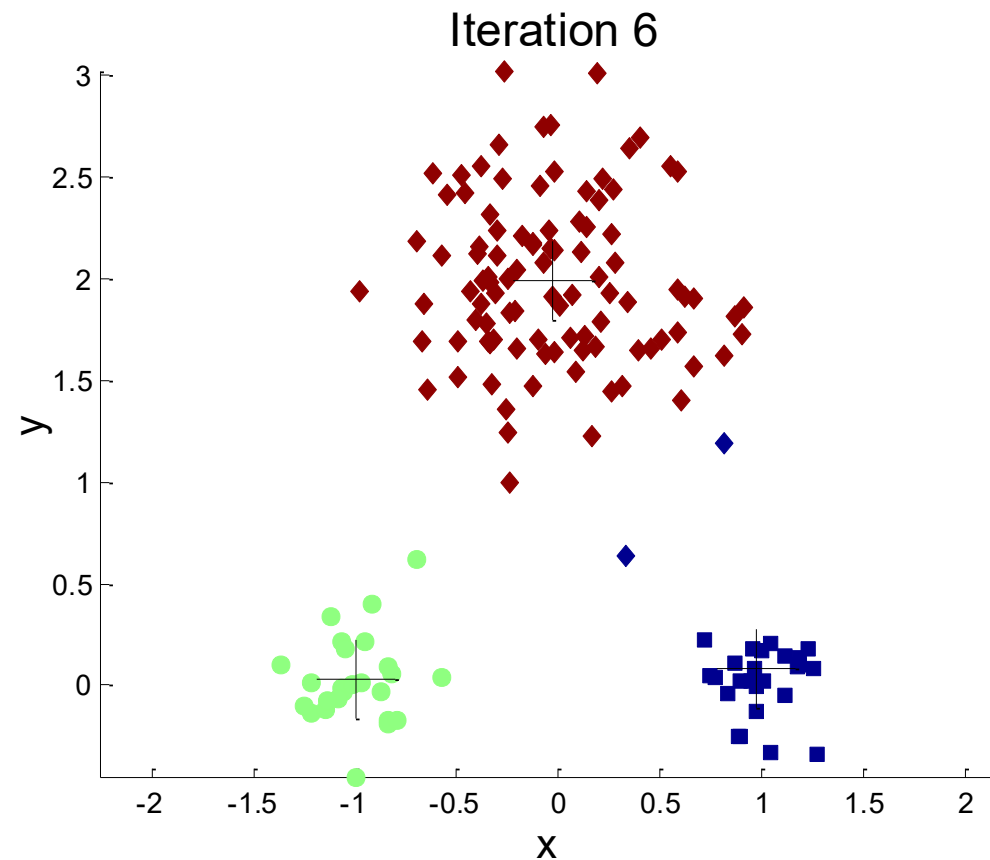


Optimal Clustering

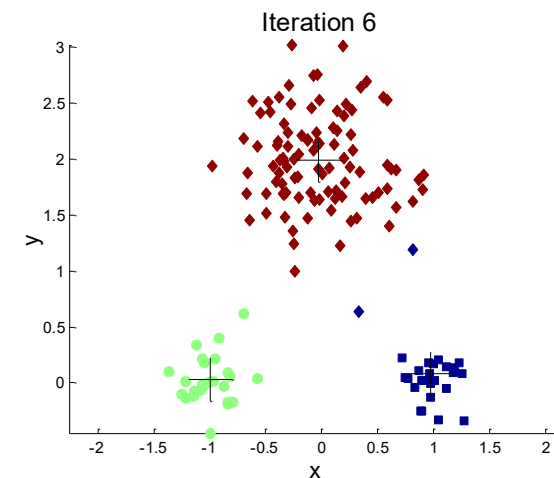
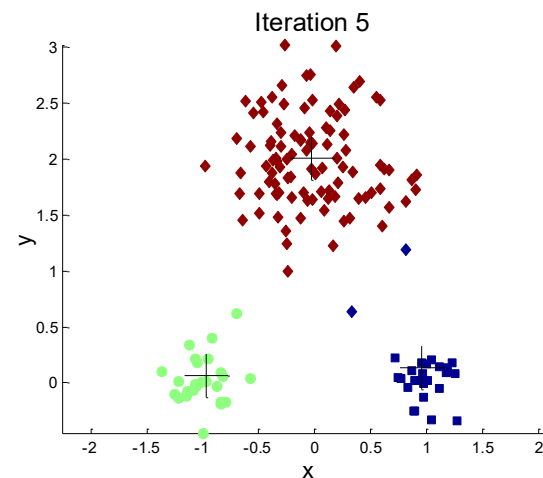
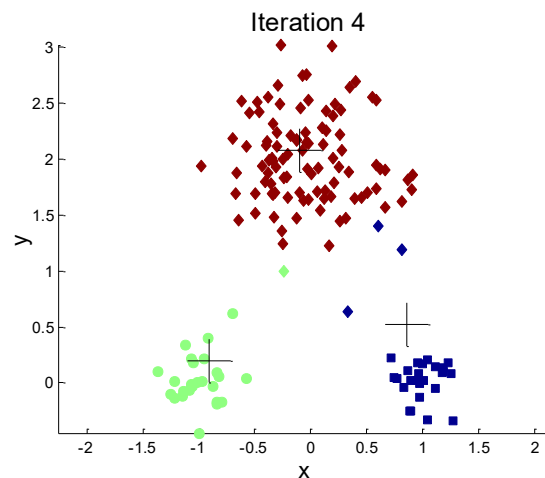
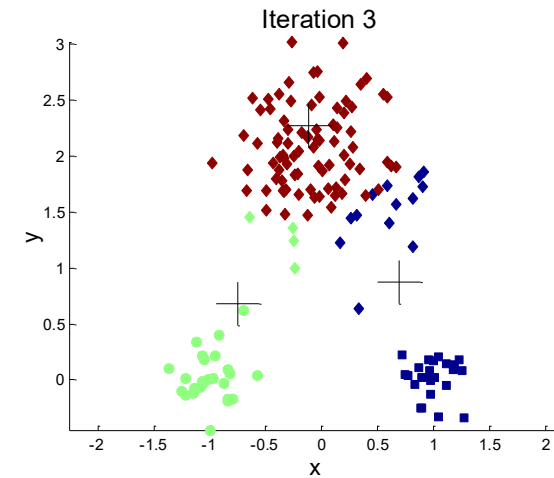
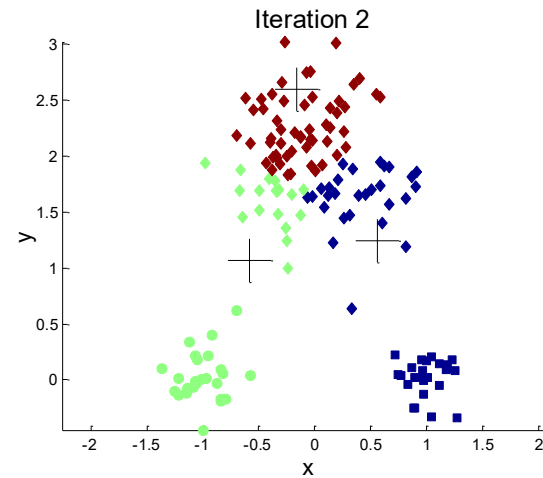
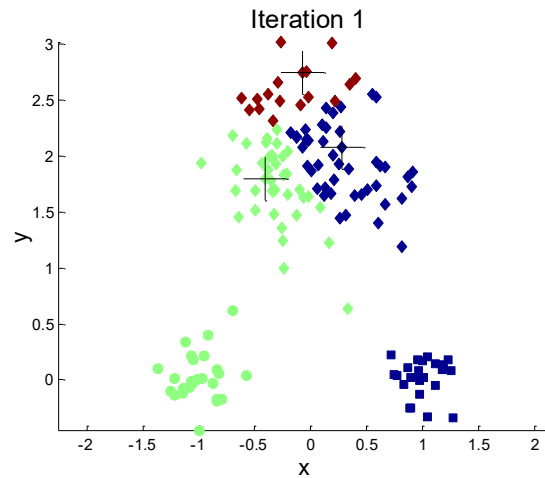


Sub-optimal Clustering

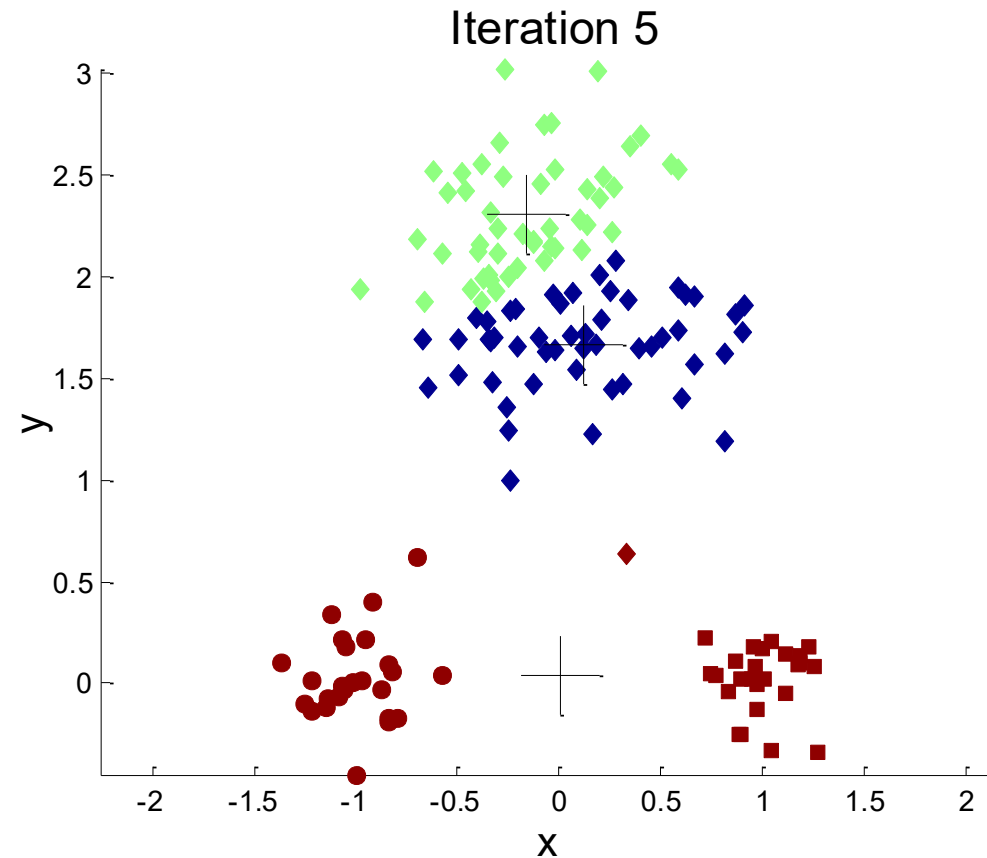
Importance of Choosing Initial Centroids



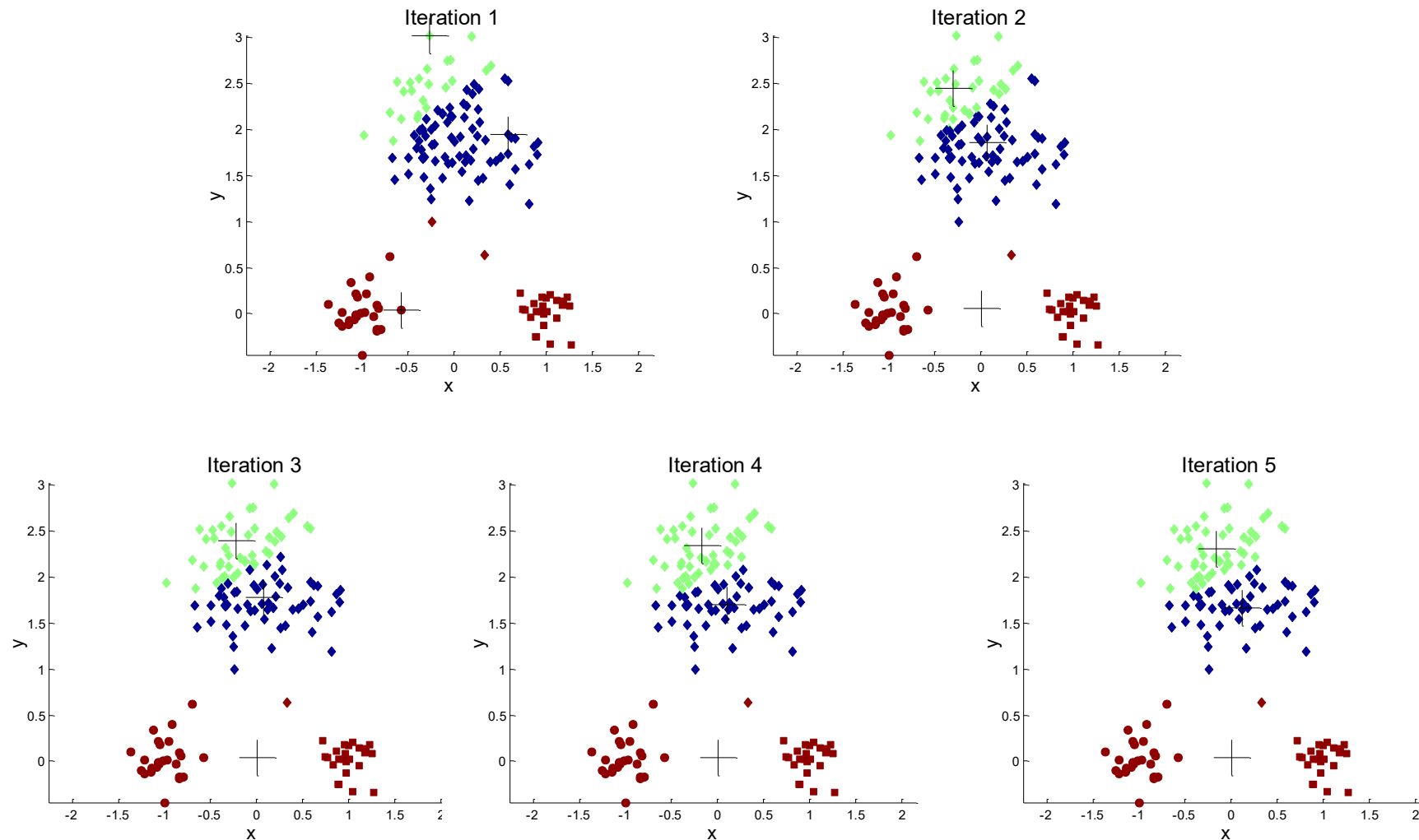
Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



Solutions to Initial Centroids Problem

- **Multiple runs**

- Helps, but probability is not on your side

- **Sample and use hierarchical clustering to determine initial centroids**

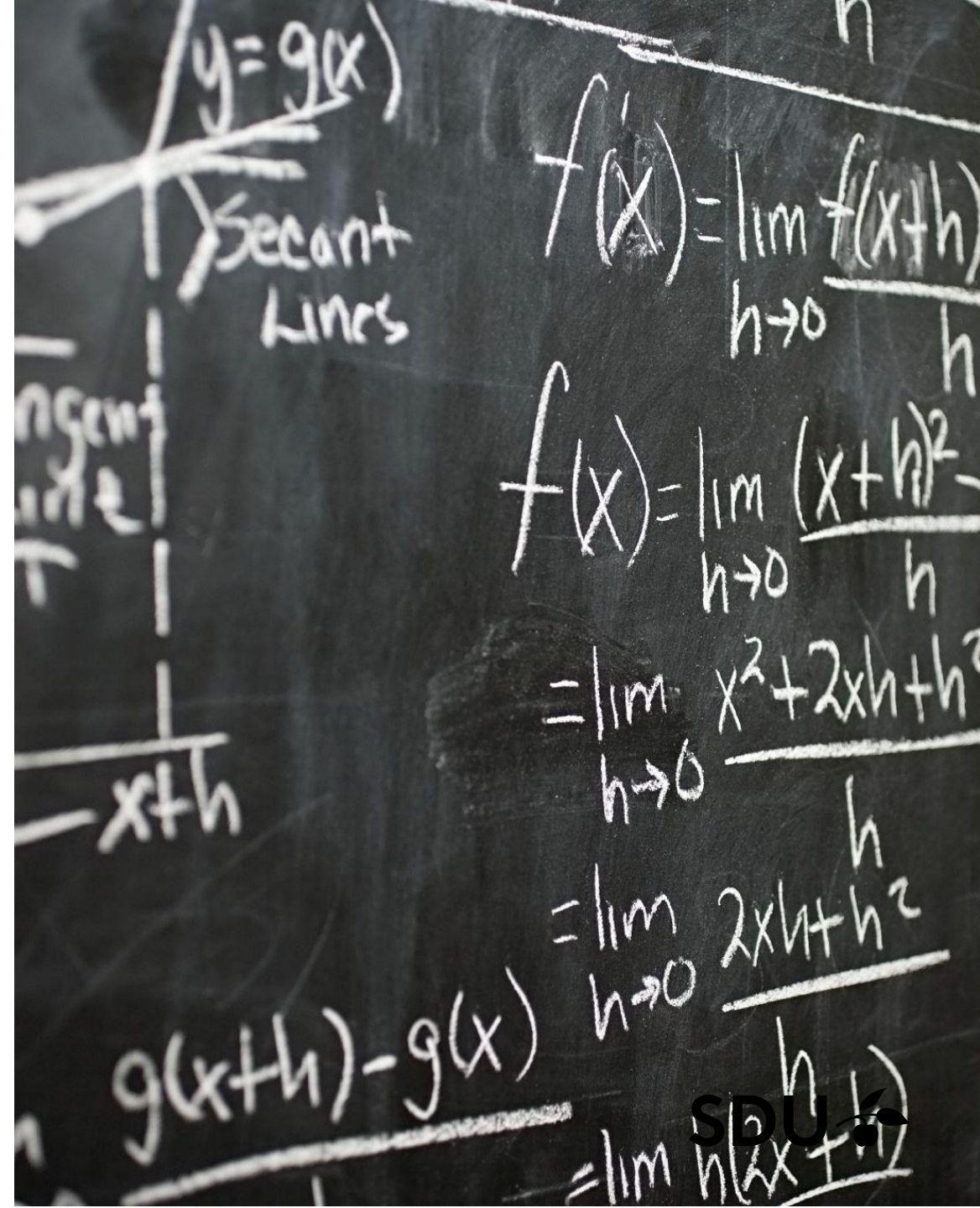
- **Select more than k initial centroids and then select among these initial centroids**

- Select most widely separated

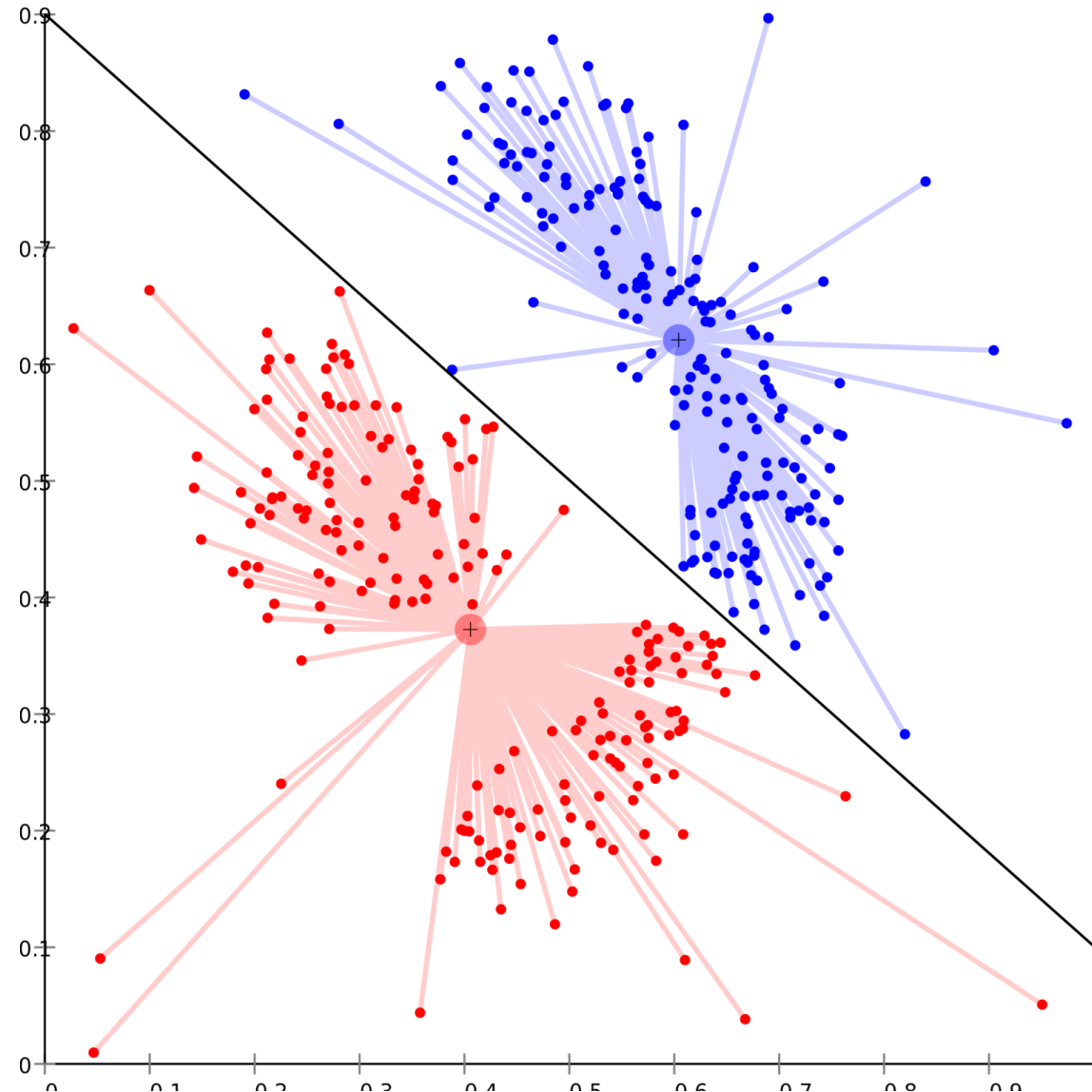
- **Postprocessing**

- **Bisecting K-means**

- Not as susceptible to initialization issues

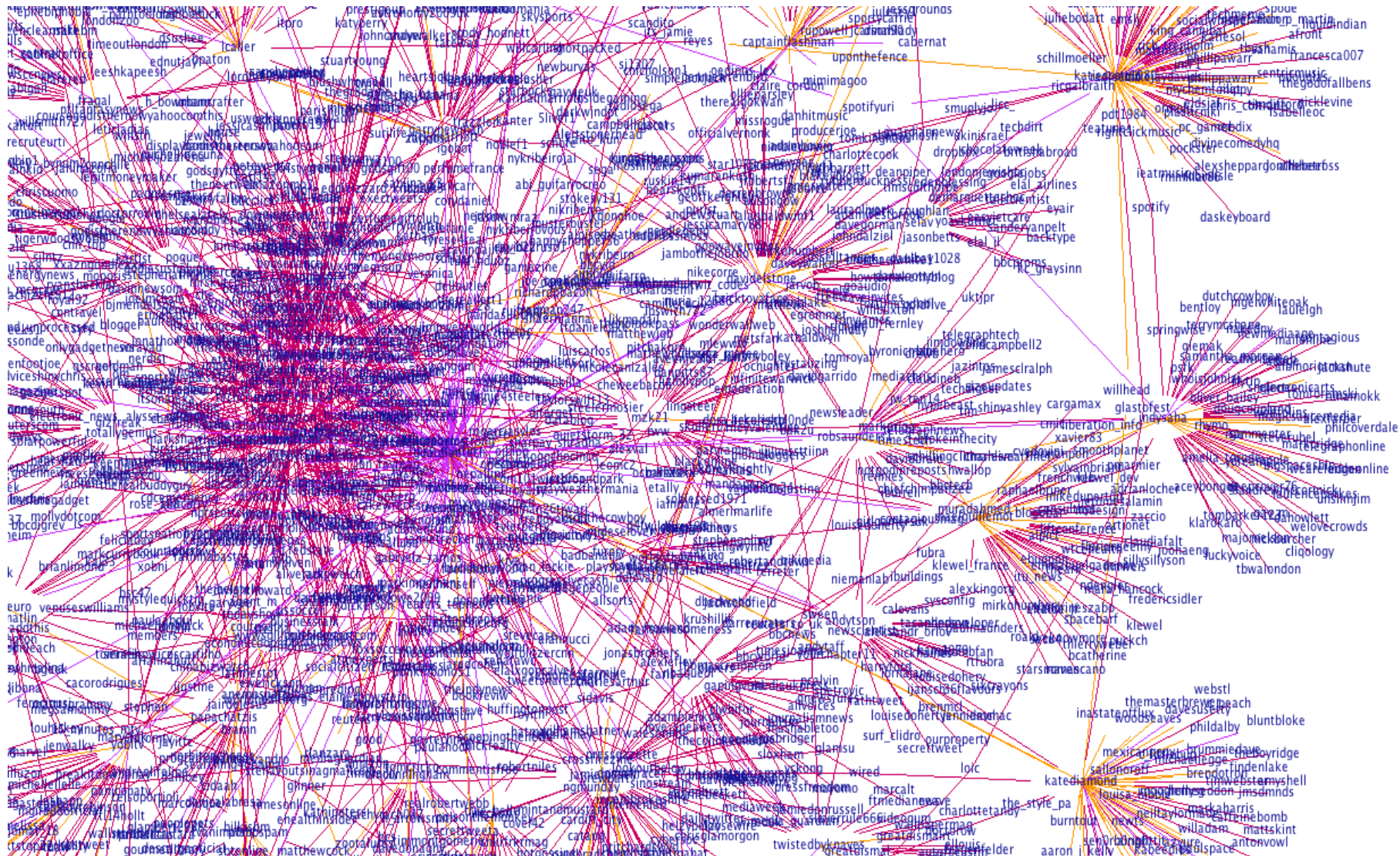


Bisecting K-Means:



Picture Ref: https://en.wikipedia.org/wiki/Cluster_analysis

What do to in high-dimensional data?



Hierarchical Clustering

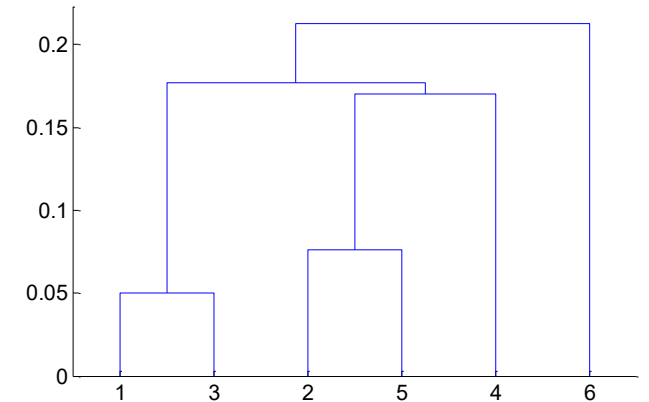
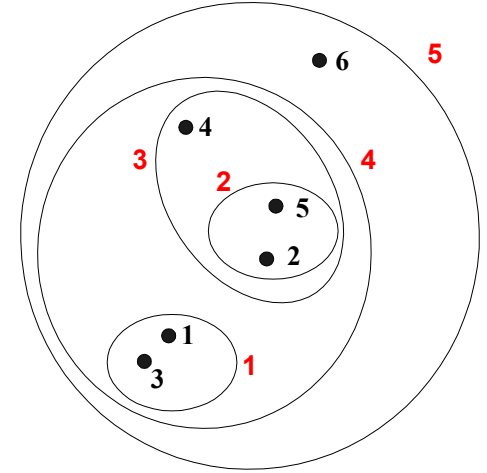
Produces a set of **nested clusters** organized as a **hierarchical tree**

Can be **visualized as a dendrogram**

- A tree like diagram that records the sequences of merges or splits

Do not have to **assume** any particular **number of clusters**

- Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level



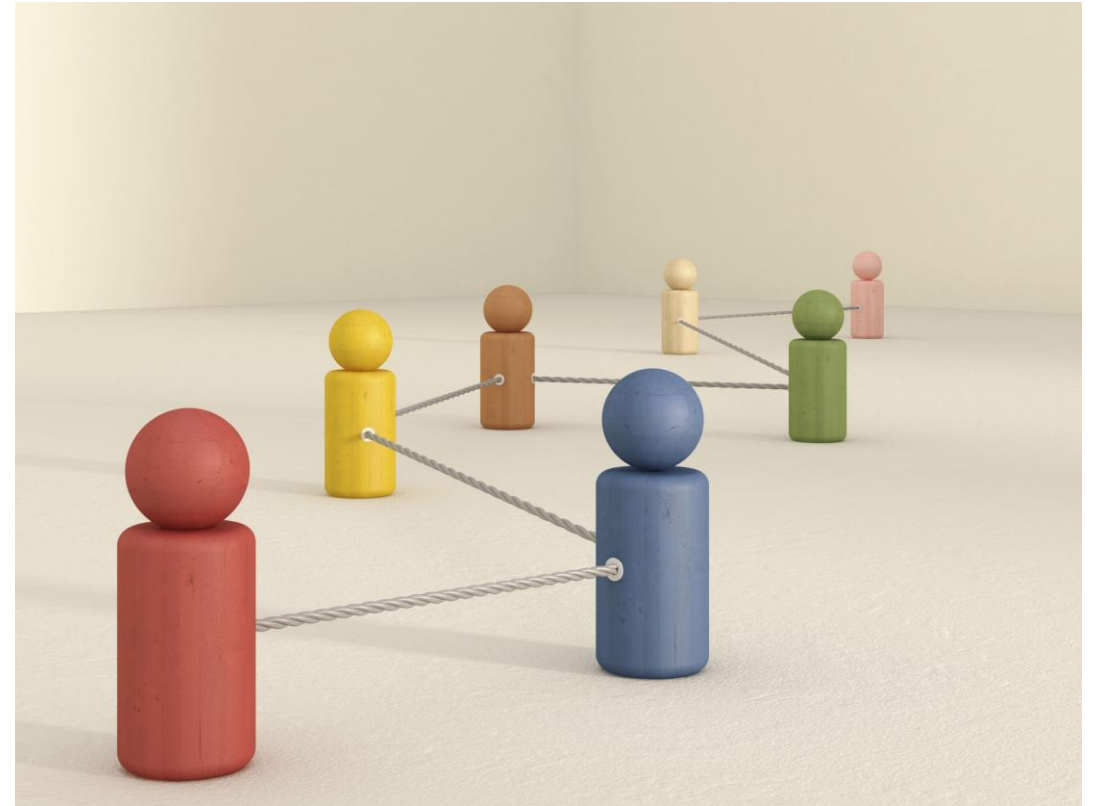
Hierarchical Clustering

Two main types of hierarchical clustering

- Agglomerative:
 - Start with the points as individual clusters
 - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
- Divisive:
 - Start with one, all-inclusive cluster
 - At each step, split a cluster until each cluster contains a point (or there are k clusters)

Traditional hierarchical algorithms use a similarity or distance matrix

- Merge or split one cluster at a time



Agglomerative Clustering Algorithm

More popular hierarchical clustering technique.

Basic algorithm is straightforward:

1. Compute the proximity matrix
2. Let each data point be a cluster
3. **Repeat**
4. Merge the two closest clusters
5. Update the proximity matrix
6. **Until** only a single cluster remains

Key operation is the computation of the proximity of two clusters

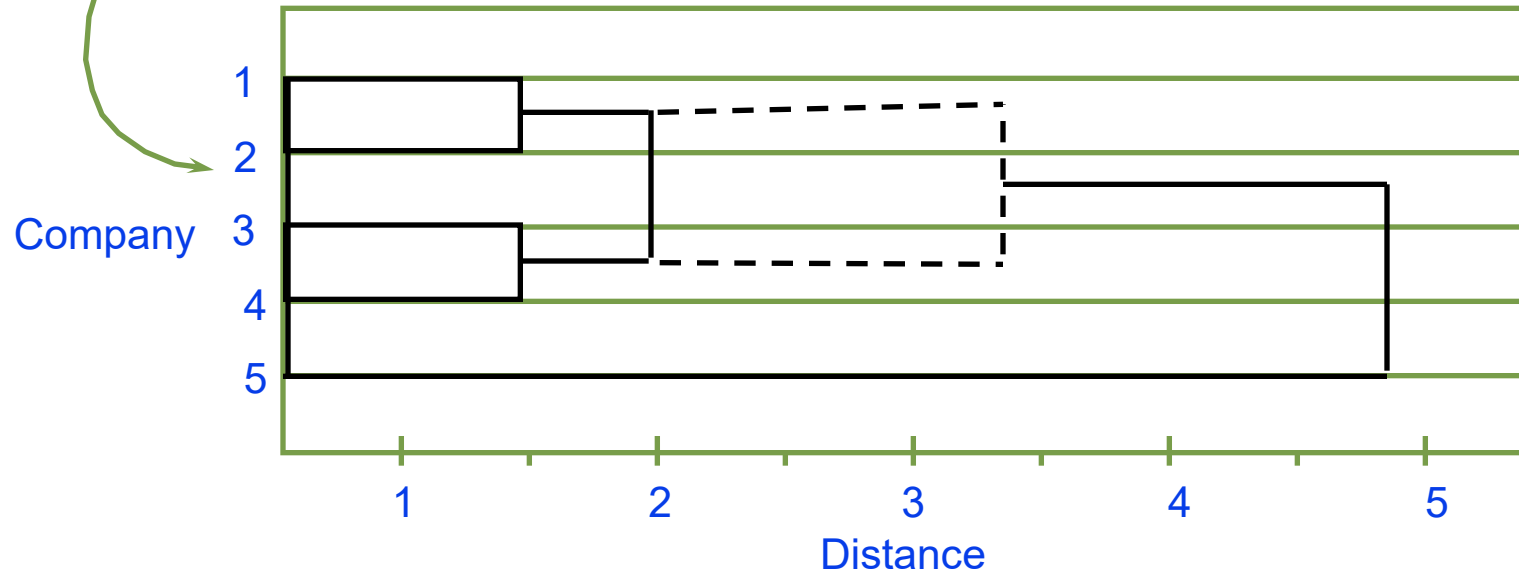
- Different approaches to defining the distance between clusters distinguish the different algorithms

Cluster Example

Distance Matrix

	Co#1	Co#2	Co#3	Co#4	Co#5
Company #1	0.00				
Company #2	1.49	0.00			
Company #3	3.42	2.29	0.00		
Company #4	1.81	1.99	1.48	0.00	
Company #5	5.05	4.82	4.94	4.83	0.00

Resulting
Dendrogram

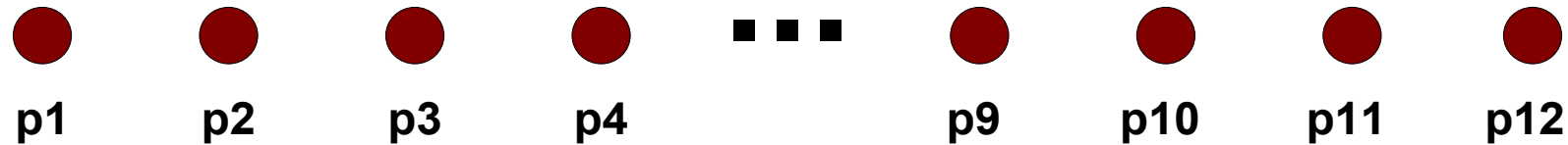


Starting Situation

Start with clusters of individual points and a proximity matrix

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

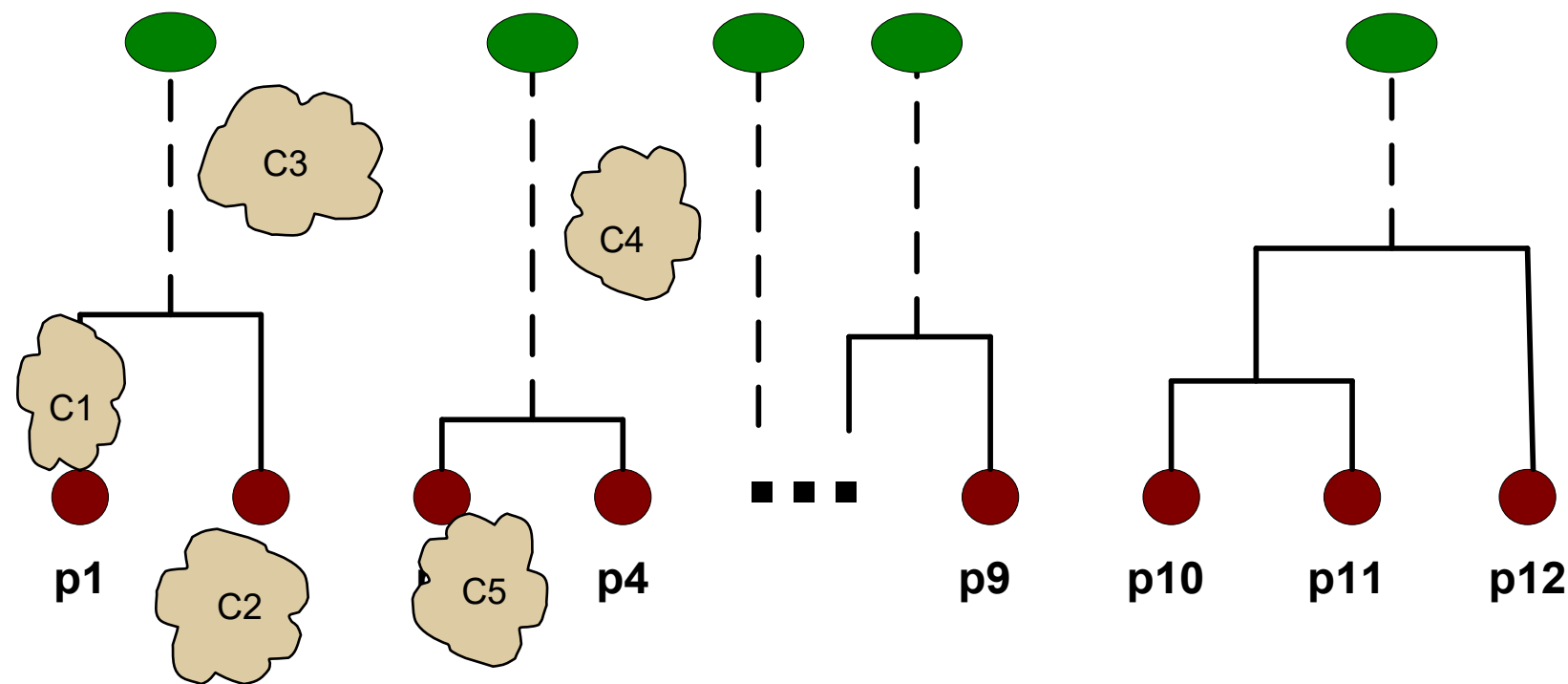


Intermediate Situation

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

After some merging steps, we have some clusters

Proximity Matrix

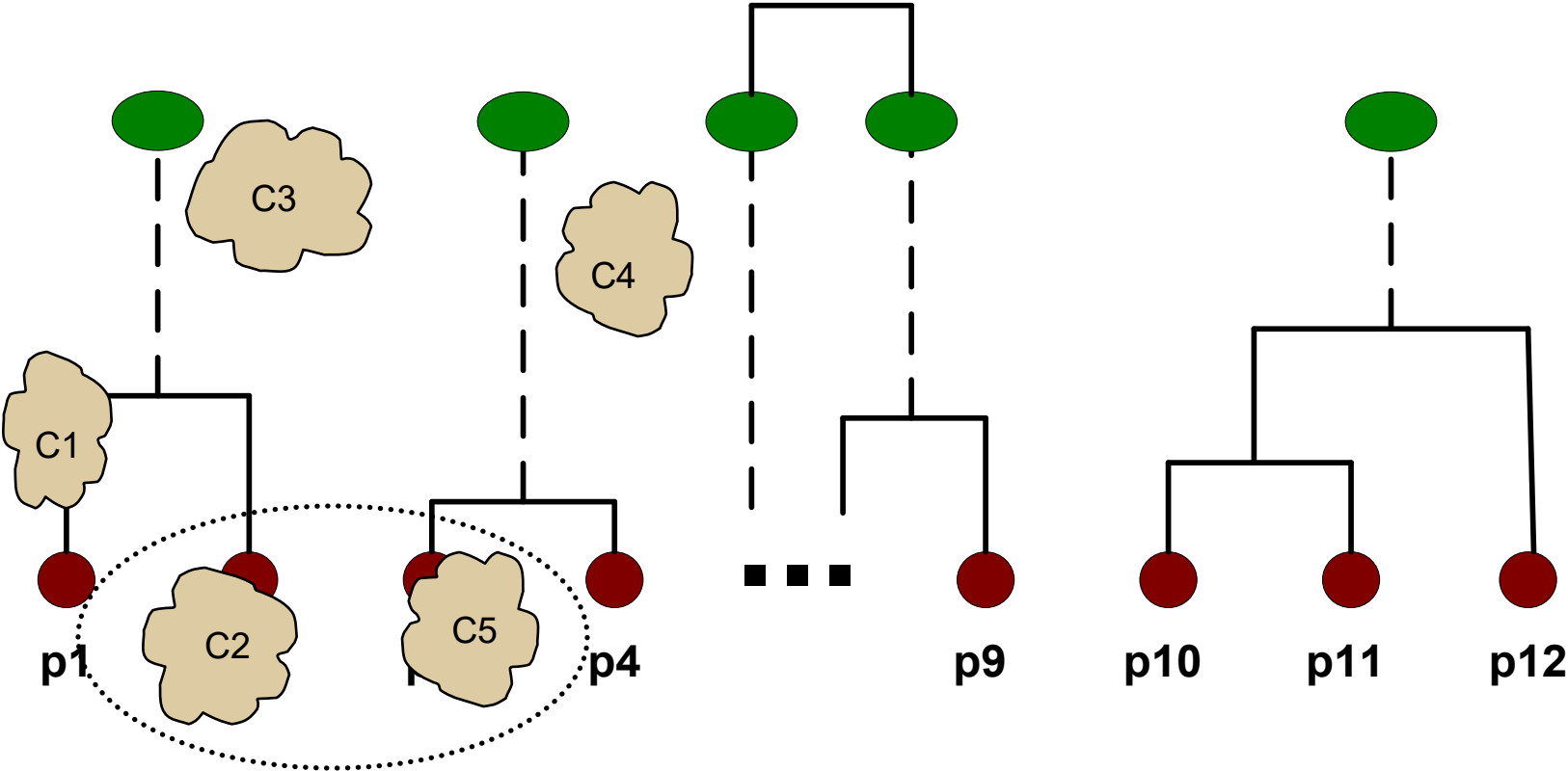


Intermediate Situation

We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

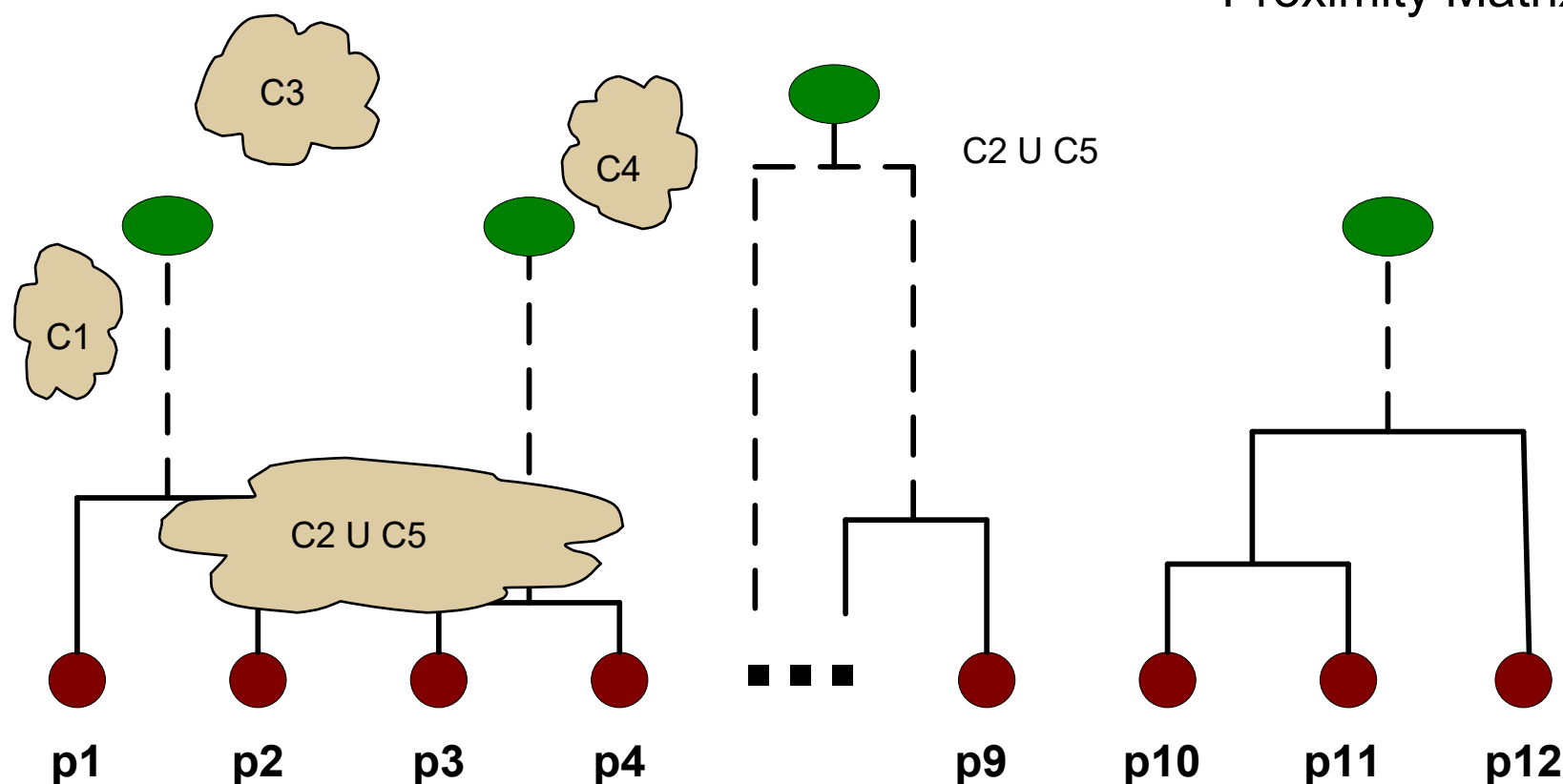


After Merging

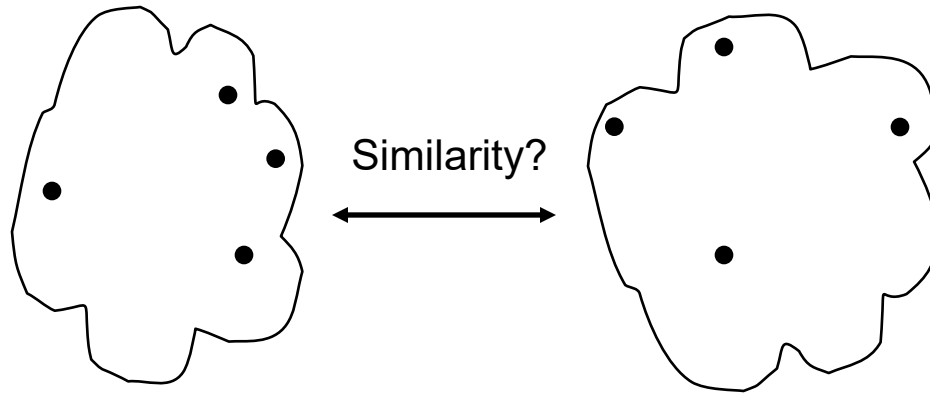
The question is “How do we update the proximity matrix?”

	C1	C2 U C5	C3	C4
C1		?		
	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



How to Define Inter-Cluster Similarity

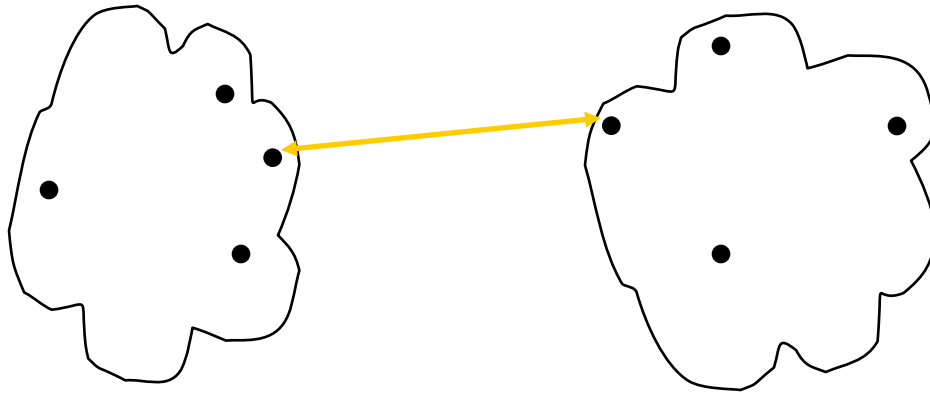


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

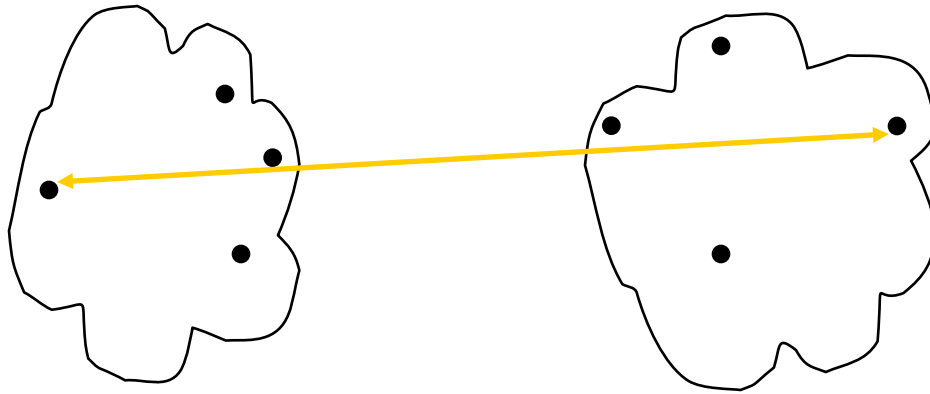


- ❑ MIN
- ❑ MAX
- ❑ Group Average
- ❑ Distance Between Centroids
- ❑ Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

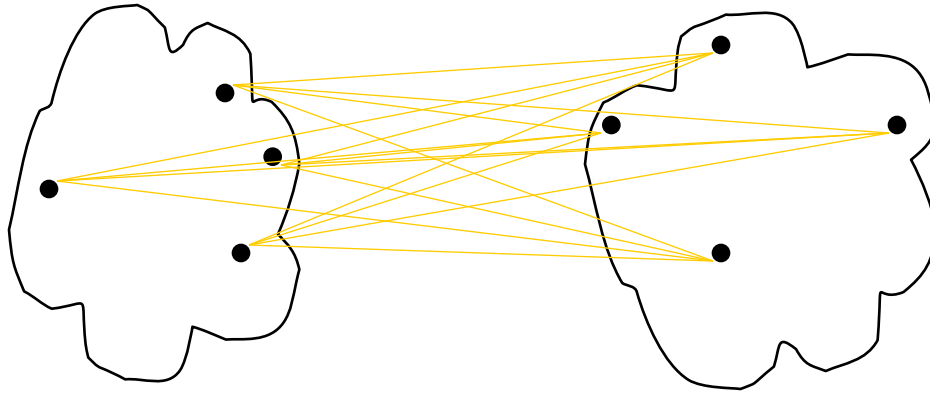


- ❑ MIN
- ❑ **MAX**
- ❑ Group Average
- ❑ Distance Between Centroids
- ❑ Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

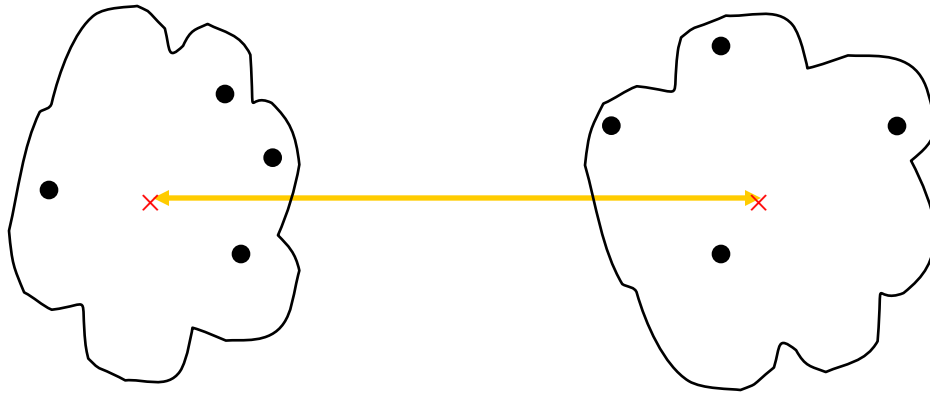


- ❑ MIN
- ❑ MAX
- ❑ **Group Average**
- ❑ Distance Between Centroids
- ❑ Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity



- ❑ MIN
- ❑ MAX
- ❑ Group Average
- ❑ Distance Between Centroids
- ❑ Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

Cluster Similarity: Ward's Method

Similarity of two clusters is based on the increase in squared error when two clusters are merged

- Similar to group average if distance between points is distance squared

Less susceptible to noise and outliers

Biased towards globular clusters

Hierarchical analogue of K-means

- Can be used to initialize K-means

Hierarchical Clustering: Problems and Limitations

Computational complexity in time and space

Once a decision is made to combine two clusters, it cannot be undone

No objective function is directly minimized

Different schemes have problems with one or more of the following:

- Sensitivity to noise and outliers
- Difficulty handling different sized clusters and convex shapes
- Breaking large clusters



DBSCAN: Density-Based Clustering

DBSCAN is a Density-Based Clustering algorithm

Remark: In density based clustering we **partition** points into **dense regions** separated by **not-so-dense regions**.

Important Questions:

- How do we measure density?
- What is a dense region?

DBSCAN:

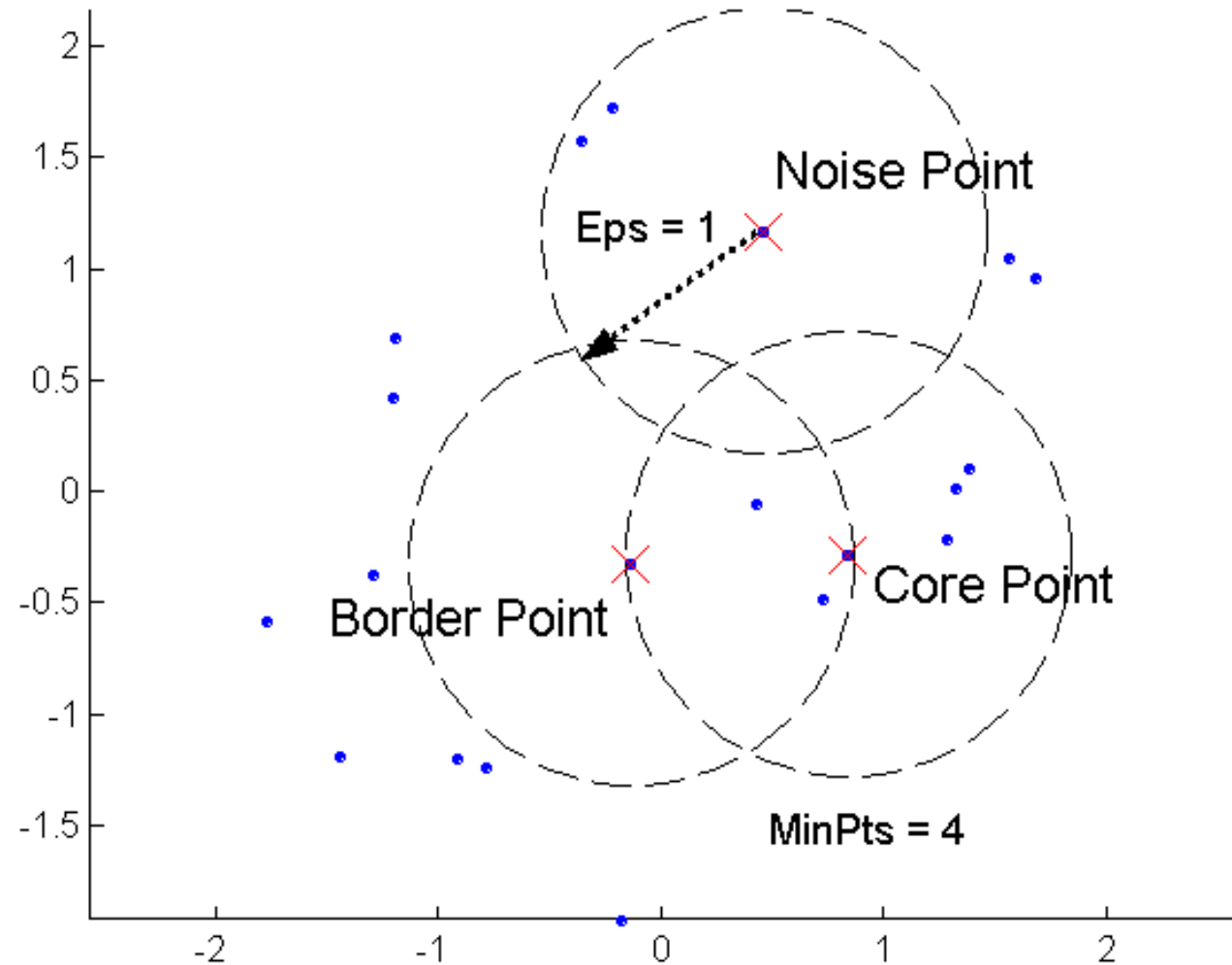
- **Density at point p :** number of points within a circle of radius Eps
- **Dense Region:** A circle of radius Eps that contains at least $MinPts$ points

DBSCAN

Characterization of points

- A point is a **core point** if it has more than a specified number of points (**MinPts**) within **Eps**
 - These points belong in a **dense region** and are at the **interior** of a cluster
- A **border point** has fewer than **MinPts** within **Eps**, but is in the neighborhood of a **core** point.
- A **noise point** is any point that is not a core point or a border point.

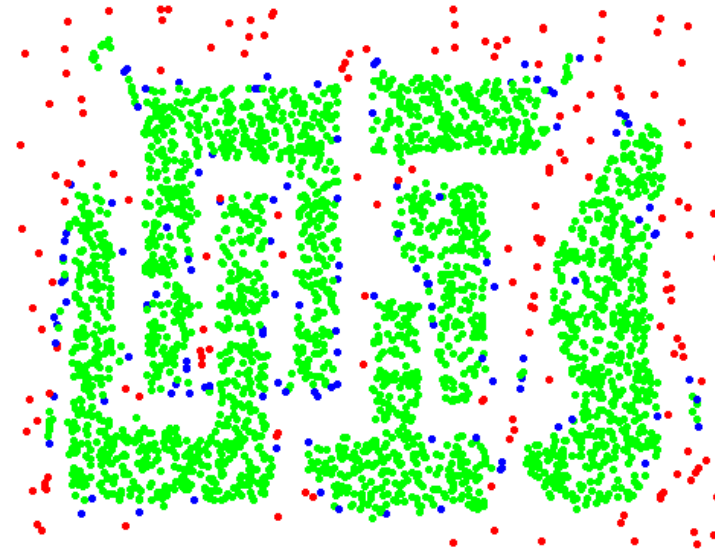
DBSCAN: Core, Border, and Noise Points



DBSCAN: Core, Border and Noise Points



Original Points



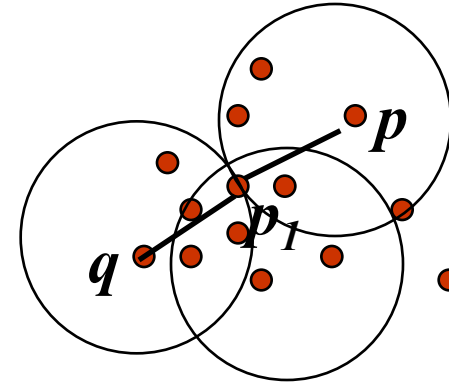
Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

Density-Connected points

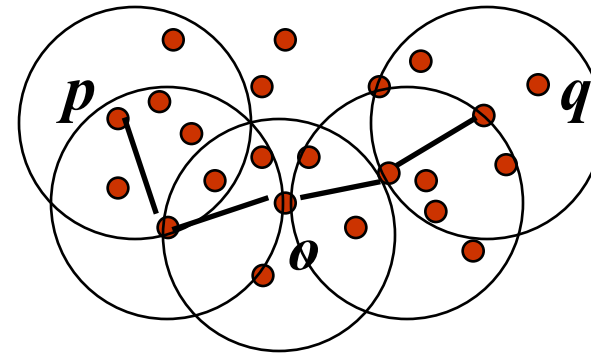
Density edge

- We place an **edge** between two core points **q** and **p** if they are within distance **Eps**.



Density-connected

- A point **p** is **density-connected** to a point **q** if there is a **path of edges** from **p** to **q**



DBSCAN Algorithm

1. Label points as **core**, **border** and **noise**
2. Eliminate **noise** points
3. For every **core** point **p** that has not been assigned to a cluster
 - Create a new cluster with the point **p** and all the points that are **density-connected** to **p**.
4. Assign border points to the cluster of the closest core point.

DBSCAN: Determining Eps and MinPts

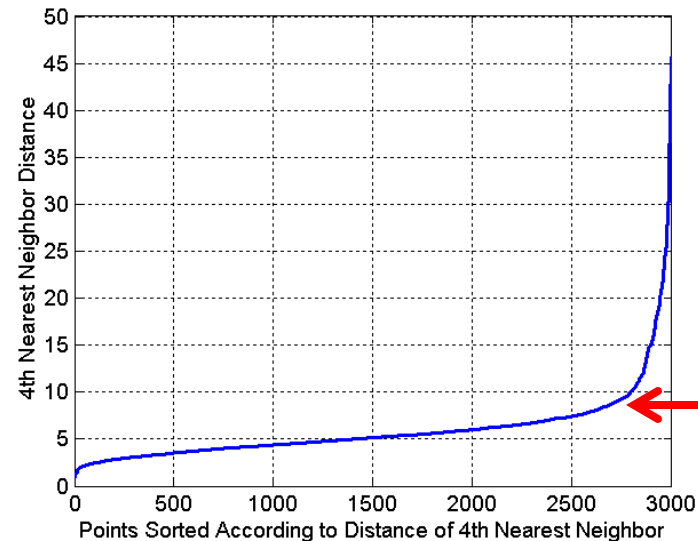
Idea is that for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance

Noise points have the k^{th} nearest neighbor at farther distance

So, plot sorted distance of every point to its k^{th} nearest neighbor

Find the distance d where there is a “knee” in the curve

- $\text{Eps} = d$, $\text{MinPts} = k$

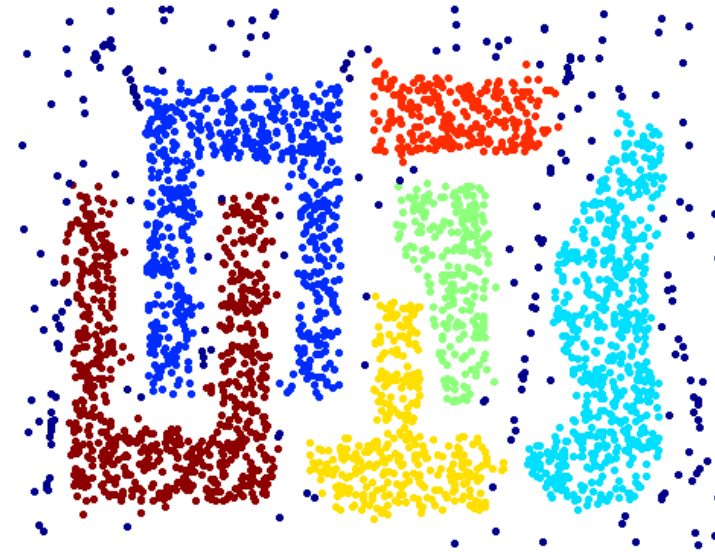


Eps ~ 7-10
MinPts = 4

When DBSCAN Works Well



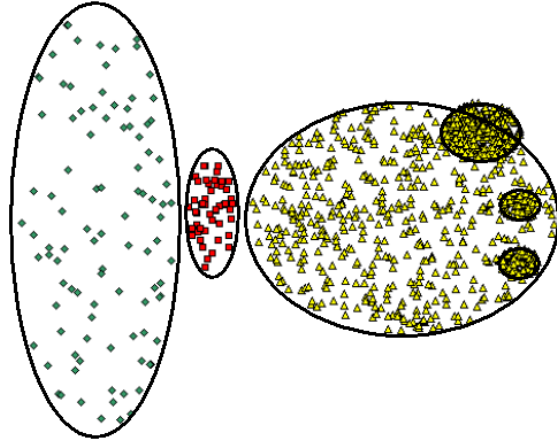
Original Points



Clusters

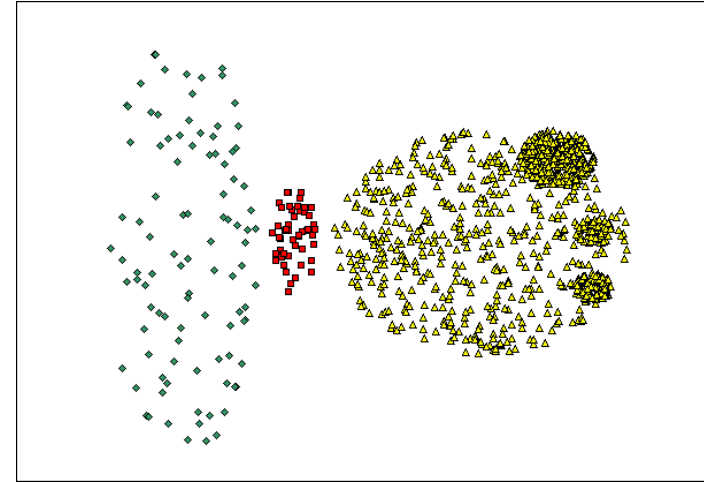
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

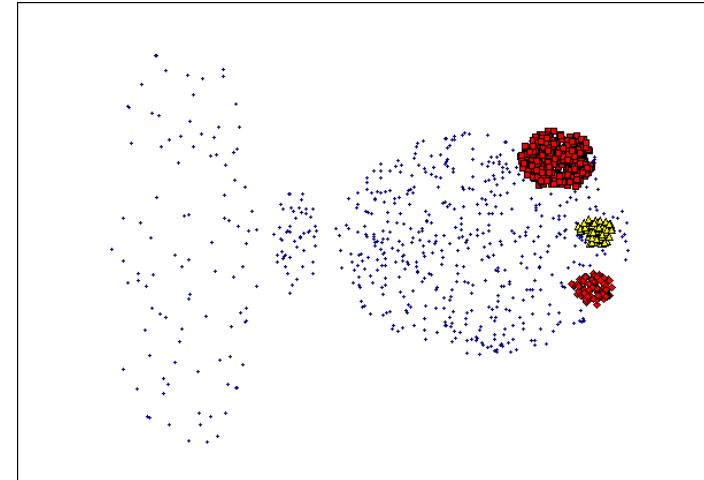


Original Points

- Sensitive to Parameters
- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

Cluster Validity



For cluster analysis, the question is how to evaluate the “goodness” of the resulting clusters?



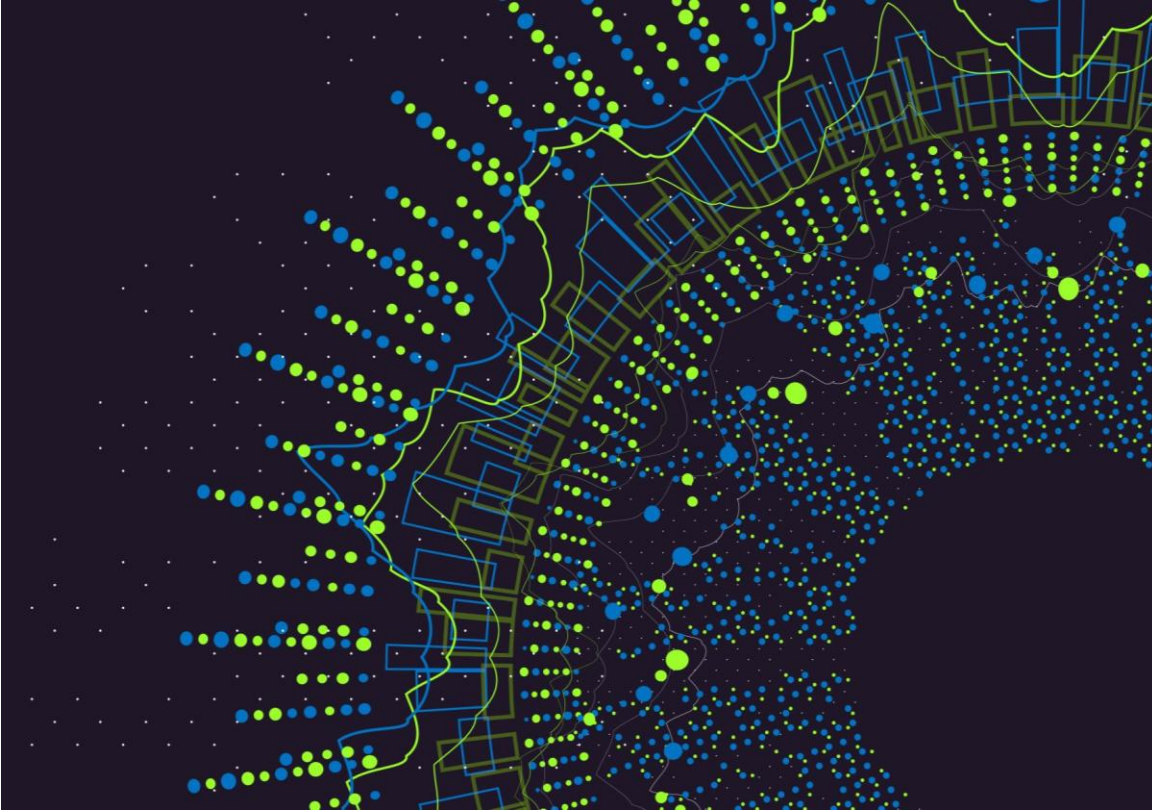
But “clusters are in the eye of the beholder”!



Then why do we want to evaluate them?

- To avoid finding patterns in noise
- To compare clustering algorithms
- To compare two sets of clusters
- To compare two clusters

Different Aspects of Cluster Validation



1. **Determining the clustering tendency of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.**
2. **Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.**
3. **Evaluating how well the results of a cluster analysis fit the data *without* reference to external information.**

- Use only the data

4. **Comparing the results of two different sets of cluster analyses to determine which is better.**
5. **Determining the 'correct' number of clusters.**

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.

Final Comments on Cluster Validity

“The validation of clustering structures is the most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage.”

Algorithms for Clustering Data, Jain and Dubes

**Thanks for your
attention**